

Data Dictionary

Table & Column Definitions — Airline Flight Delay Analysis

Data Source	Maven Analytics — Airline Flight Delays (U.S. DOT, 2015)
Tables	flights, airlines, airports
Version	1.0
Date	August 2026

1. Overview

The dataset is delivered as three CSV files that together form a simple star schema: one large fact table (`flights`) and two small reference tables (`airlines` , `airports`). This section documents every field in each table, its data type, whether it can be blank/null, and its business meaning.

2. Table: airlines

Grain: one row per airline. **Row count:** ~14 major U.S. carriers. **Role:** lookup/dimension table joined to `flights.AIRLINE`.

Column	Data Type	Key	Nullable	Description	Example
<code>IATA_CODE</code>	<code>varchar(2)</code>	PK	No	Two-letter IATA airline code, unique identifier for the carrier.	AA
<code>AIRLINE</code>	<code>varchar(100)</code>	—	No	Full legal/trading name of the airline.	American Airlines Inc.

3. Table: airports

Grain: one row per airport. **Row count:** ~322 U.S. airports. **Role:** lookup/dimension table, joined twice to the flights table — once for the origin airport and once for the destination airport.

Column	Data Type	Key	Nullable	Description	Example
<code>IATA_CODE</code>	<code>varchar(3)</code>	PK	No	Three-letter IATA airport code, unique identifier for the airport.	JFK
<code>AIRPORT</code>	<code>varchar(150)</code>	—	No	Full name of the airport.	John F. Kennedy International Airport
<code>CITY</code>	<code>varchar(100)</code>	—	No	City the airport serves.	New York
<code>STATE</code>	<code>varchar(2)</code>	—	No	Two-letter U.S. state abbreviation.	NY
<code>COUNTRY</code>	<code>varchar(50)</code>	—	No	Country the airport is located in.	USA
<code>LATITUDE</code>	<code>float</code>	—	Yes (a few rows)	Geographic latitude, used for map visuals.	40.6398
<code>LONGITUDE</code>	<code>float</code>	—	Yes (a few rows)	Geographic longitude, used for map visuals.	-73.7789

4. Table: flights (Fact Table)

Grain: one row per scheduled flight operation on a given date. **Row count:** 5,000,000+ for calendar year 2015. **Role:** central fact table containing all time, delay, and status measures; connects to `airlines` and `airports` via foreign keys.

4.1 Identifier & Date Fields

Column	Data Type	Key	Nullable	Description	Example
<code>YEAR</code>	<code>int</code>	—	No	Calendar year of the flight. Constant 2015 for this dataset.	2015
<code>MONTH</code>	<code>int (1-12)</code>	—	No	Calendar month of the flight.	7
<code>DAY</code>	<code>int (1-31)</code>	—	No	Day of the month.	15
<code>DAY_OF_WEEK</code>	<code>int (1-7)</code>	—	No	Day of week, 1 = Monday through 7 = Sunday.	3
<code>AIRLINE</code>	<code>varchar(2)</code>	FK → <code>airlines.IATA_CODE</code>	No	IATA code of the operating airline.	DL
<code>FLIGHT_NUMBER</code>	<code>varchar(10)</code>	—	No	Airline-assigned flight number.	1545
<code>TAIL_NUMBER</code>	<code>varchar(10)</code>	—	Yes	Registration number of the specific aircraft that operated the flight.	N3730B

ORIGIN_AIRPORT	varchar(3)	FK → airports.IATA_CODE	No	IATA code of the departure airport.	ATL
DESTINATION_AIRPORT	varchar(3)	FK → airports.IATA_CODE	No	IATA code of the arrival airport.	ORD

4.2 Departure Fields

Column	Data Type	Nullable	Description	Example
SCHEDULED_DEPARTURE	time (HHMM)	No	Planned local departure time.	0805
DEPARTURE_TIME	time (HHMM)	Yes (blank if cancelled)	Actual gate-departure time.	0812
DEPARTURE_DELAY	int (minutes)	Yes	Minutes between scheduled and actual departure. Negative = early departure.	7
TAXI_OUT	int (minutes)	Yes	Minutes spent taxiing from gate to takeoff.	16
WHEELS_OFF	time (HHMM)	Yes	Actual time the aircraft leaves the ground.	0828

4.3 Flight Duration Fields

Column	Data Type	Nullable	Description	Example
SCHEDULED_TIME	int (minutes)	No	Planned total gate-to-gate flight duration.	150
ELAPSED_TIME	int (minutes)	Yes	Actual total gate-to-gate flight duration.	158
AIR_TIME	int (minutes)	Yes	Minutes actually airborne (wheels-off to wheels-on).	128
DISTANCE	int (miles)	No	Great-circle distance flown between origin and destination.	606

4.4 Arrival Fields

Column	Data Type	Nullable	Description	Example
WHEELS_ON	time (HHMM)	Yes	Actual time the aircraft touches down.	1036
TAXI_IN	int (minutes)	Yes	Minutes spent taxiing from landing to the gate.	9
SCHEDULED_ARRIVAL	time (HHMM)	No	Planned local arrival time.	1035
ARRIVAL_TIME	time (HHMM)	Yes	Actual gate-arrival time.	1045
ARRIVAL_DELAY	int (minutes)	Yes	Minutes between scheduled and actual arrival. Negative = early arrival.	10

4.5 Status & Cancellation Fields

Column	Data Type	Nullable	Description	Example
DIVERTED	boolean (0/1)	No	1 if the flight landed at an airport other than its scheduled destination.	0
CANCELLED	boolean (0/1)	No	1 if the flight was cancelled and never operated.	0
CANCELLATION_REASON	varchar(1)	Yes (only if CANCELLED=1)	A = Airline/Carrier, B = Weather, C = National Air System, D = Security.	B

4.6 Delay Cause Fields (populated only when ARRIVAL_DELAY ≥ 15 minutes)

Column	Data Type	Nullable	Description	Example
AIR_SYSTEM_DELAY	int (minutes)	Yes	Delay minutes attributed to the National Air System (ATC, airspace congestion).	0
SECURITY_DELAY	int	Yes	Delay minutes attributed to security incidents.	0

	(minutes)			
AIRLINE_DELAY	int (minutes)	Yes	Delay minutes attributed to the airline/carrier itself.	12
LATE_AIRCRAFT_DELAY	int (minutes)	Yes	Delay minutes caused by the incoming aircraft arriving late.	0
WEATHER_DELAY	int (minutes)	Yes	Delay minutes attributed directly to weather.	0

Data Quality Note: Delay-cause columns (Section 4.6) and several time/duration columns are intentionally blank for flights that were not delayed 15+ minutes, or that were cancelled/diverted before reaching that milestone. These blanks represent "not applicable," not missing/dirty data, and should be treated as zero rather than excluded when aggregating total delay minutes.

5. Field Naming Conventions

- All column names use `UPPER_SNAKE_CASE`, matching the original DOT/Kaggle export.
- Time fields (e.g. `SCHEDULED_DEPARTURE`) are stored as 4-digit 24-hour local time integers (HHMM) and should be parsed into proper time values during data preparation.
- All "_DELAY" minute fields can be negative (early) or positive (late); only positive values count toward delay KPI totals per DOT convention.